

# LEO LEONG

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Currently serving National Service, available for full-time work from 2nd August 2026

## SUMMARY

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Diploma graduate in Cybersecurity & Digital Forensics from Temasek Polytechnic with hands-on experience in software development, networking, and cybersecurity. Built and deployed a live IoT air quality monitoring platform on Azure as a Final Year Project. Completed an internship at Trend Micro developing POC security demonstrations and automation tooling for enterprise clients. Currently seeking an entry-level software engineer role.

## SKILLS

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**Programming:** Python, Java, JavaScript, TypeScript, PHP, HTML/CSS, SQL

**Frameworks & Tools:** Next.js, React, Express.js, Sequelize ORM, MQTT.js, Nodemailer, SendGrid, JWT, bcrypt, Git

**Cloud & Databases:** Microsoft Azure, PostgreSQL, MySQL

**Cybersecurity Tools:** Burp Suite, Nmap, Metasploit, Kali Linux, Wireshark

**Data & ML:** scikit-learn, pandas, Streamlit, Power BI

**Systems:** Linux (Red Hat), Nginx, Android Studio

**Certificates:** AWS Academy Cloud Foundations

## WORK EXPERIENCE

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### Technical Intern

April 2023 – September 2023

*Trend Micro*

- Developed POC documentation for Trend Micro security products by deploying controlled VM environments, injecting test malware, and validating detection, logging, and remediation behaviour. Produced reports adopted by Sales Engineers in enterprise client presentations
- Supported Sales Engineers and Technical Account Managers in delivering cybersecurity solutions to clients, coordinating technical requests across departments
- Developed a Python automation tool that extracted client data (e.g., name, email, and product details) from Excel spreadsheets and populated a standardized email template with customer-specific information for outreach campaigns. Independently designed, tested, and debugged the solution, eliminating manual one-by-one email distribution for the Sales Engineering team.

### Vehicle Commander, Terrex ICV

2025 – Present

*Singapore Armed Forces*

- Leads a vehicle crew and dismounted infantry squad, managing real-time battlefield situational awareness via the Battlefield Management System (BMS) and coordinating mission execution under operational pressure
- Responsible for crew readiness, mission briefings, and tactical decision-making in high-stakes, time-critical environments, developing leadership and composure that complements technical work

## PROJECTS

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### EcoSaver — IoT Air Quality Monitoring Platform | *Node.js, Express, MySQL, Azure, MQTT, EJS, Nginx*

- Designed and built a full-stack air quality monitoring web platform integrating live IoT sensor data via MQTT, deployed on Microsoft Azure with a custom domain and SSL/TLS certificates
- Developed a consumer-facing website displaying real-time and historical air quality metrics (PSI, NO2, SO2, CO, O3, Temperature, Humidity) with daily and weekly data visualisation charts
- Built a separate admin management portal with dashboards for server uptime, database status, CPU monitoring, user management, sensor management, and API log tracking
- Designed and documented a RESTful API with token-based authentication, enabling third-party access to sensor data
- Configured Linux server environment including Nginx reverse proxy routing, systemd service management, and Let's Encrypt SSL certificate integration

### Leo's Cafe — Full-Stack Café | *Node.js, Express, PostgreSQL, Sequelize ORM, JWT, bcrypt, SendGrid, QR Code, EJS*

- Built a production-grade café platform with customer-facing ordering, a tiered loyalty programme, QR code redemptions, and table reservation management

- Implemented security-first authentication: JWT stored in httpOnly cookies, bcrypt password hashing, SendGrid email verification before account activation, and rate limiting on all auth endpoints
- Designed extensible PostgreSQL schema with Sequelize ORM; applied input validation and output sanitisation across all API routes

#### **HDB Resale Price Predictor** | *Python, pandas, scikit-learn, Random Forest, Streamlit*

- Built an end-to-end ML pipeline — data cleaning, feature engineering, and Random Forest model training on 8000 HDB resale transactions, achieving  $R^2=0.843$  and MAE of S\$68,844
- Deployed an interactive Streamlit web app allowing users to input flat details and receive instant price estimates with 5-year projections

#### **Penetration Testing Engagement** | *Nmap, Burp Suite, Metasploit, Kali Linux, Wireshark*

- Conducted a simulated professional penetration testing engagement applying the Cyber Kill Chain methodology to systematically identify vulnerabilities across a target network
- Used Nmap for network reconnaissance, Burp Suite for web application testing, and Metasploit for exploitation; produced a remediation report with CVSS-scored findings and recommended countermeasures

#### **Company Network Infrastructure Design** | *Cisco Packet Tracer, VLAN, IP Subnetting*

- Designed and configured WAN connectivity between Southeast Asia offices and overseas cloud facilities for a simulated MNC, with VLAN segmentation and IP subnetting across all sites

## **EDUCATION**

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#### **Diploma in Cybersecurity & Digital Forensics** | Temasek Polytechnic

*April 2021 – May 2024*

- Admitted into the Director's List for AY2021/2022 for excellent academic performance
- Recipient of Edusave Certificate of Academic Achievement, 2022
- Achieved Distinctions in Network Technology and Coding & Development Project

#### **Cambridge GCE O-Level** | Pasir Ris Secondary School

*2016 – 2020*

- Recipient of Edusave Scholarship (2018, 2020), Edusave Certificate of Academic Achievement (2019), and Li Kaiyan Progress Award (2020)